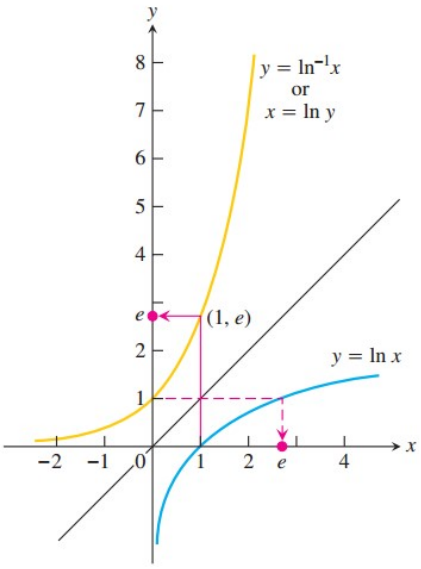
## 7.3 Exponential Functions

**DEFINITION OF THE NUMBER $e$**

$e$ is the number such that $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$

**DERIVATIVE OF THE NATURAL EXPONENTIAL FUNCTION**

$$\frac{d}{dx}(e^x) = e^x$$



**EXAMPLE 2** A line with slope $m$ passes through the origin and is tangent to the graph of $y = \ln x$. What is the value of $m$?

## Exponential Change

In modeling many real-world situations, a quantity $y$ increases or decreases at a rate proportional to its size at a given time $t$. Examples of such quantities include the size of a population, the amount of a decaying radioactive material, and the temperature difference between a hot object and its surrounding medium. Such quantities are said to undergo **exponential change**.

If the amount present at time $t = 0$ is called $y_0$, then we can find $y$ as a function of $t$ by solving the following initial value problem:

$$\text{Differential equation:} \quad \frac{dy}{dt} = ky \tag{1a}$$

$$\text{Initial condition:} \quad y = y_0 \quad \text{when} \quad t = 0. \tag{1b}$$

If $y$ is positive and increasing, then $k$ is positive, and we use Equation (1a) to say that the rate of growth is proportional to what has already been accumulated. If $y$ is positive and decreasing, then $k$ is negative, and we use Equation (1a) to say that the rate of decay is proportional to the amount still left.

In general, if $y(t)$ is the value of a quantity $y$ at time $t$ and if the rate of change of $y$ with respect to $t$ is proportional to its size $y(t)$ at any time, then

1
$$\frac{dy}{dt} = ky$$

where $k$ is a constant. Equation 1 is sometimes called the **law of natural growth** (if $k > 0$) or the **law of natural decay** (if $k < 0$). It is called a **differential equation** because it involves an unknown function $y$ and its derivative $dy/dt$.

It's not hard to think of a solution of Equation 1. This equation asks us to find a function whose derivative is a constant multiple of itself. We have met such functions in this chapter. Any exponential function of the form $y(t) = Ce^{kt}$, where $C$ is a constant, satisfies

$$y'(t) = C(ke^{kt}) = k(Ce^{kt}) = ky(t)$$

2 **THEOREM** The only solutions of the differential equation $dy/dt = ky$ are the exponential functions

$$y(t) = y(0)e^{kt}$$

**3.** A bacteria culture initially contains 100 cells and grows at a rate proportional to its size. After an hour the population has increased to 420.

(a) Find an expression for the number of bacteria after $t$ hours.

(b) Find the number of bacteria after 3 hours.

(c) Find the rate of growth after 3 hours.

(d) When will the population reach 10,000?

## The Derivative of $a^u$

To find this derivative, we start with the defining equation $a^x = e^{x \ln a}$. Then we have

$$\frac{d}{dx}a^x = \frac{d}{dx}e^{x \ln a} = e^{x \ln a} \cdot \frac{d}{dx}(x \ln a) \qquad \frac{d}{dx}e^u = e^u \frac{du}{dx}$$
$$= a^x \ln a.$$

We now see why $e^x$ is the exponential function preferred in calculus. If $a = e$, then $\ln a = 1$ and the derivative of $a^x$ simplifies to

$$\frac{d}{dx}e^x = e^x \ln e = e^x. \qquad \ln e = 1$$

With the Chain Rule, we get the following form for the derivative of the general exponential function.

If $a > 0$ and $u$ is a differentiable function of $x$, then $a^u$ is a differentiable function of $x$ and

$$\frac{d}{dx}a^u = a^u \ln a \frac{du}{dx}. \tag{3}$$

**EXAMPLE 6** We find derivatives and integrals using Equations (3) and (4).

**(a)** $\frac{d}{dx}3^x = 3^x \ln 3$ Eq. (3) with $a = 3, u = x$

**(b)** $\frac{d}{dx}3^{-x} = 3^{-x}(\ln 3)\frac{d}{dx}(-x) = -3^{-x}\ln 3$ Eq. (3) with $a = 3, u = -x$

**(c)** $\frac{d}{dx}3^{\sin x} = 3^{\sin x}(\ln 3)\frac{d}{dx}(\sin x) = 3^{\sin x}(\ln 3)\cos x$ Eq. (3) with $a = 3, u = \sin x$

**a)** $\lim_{h\to 0}\frac{3^h - 1}{h}$ **( do not use the L'Hopital's Rule)**

**2–22** Differentiate the function.

**2.** $f(x) = \ln(x^2 + 10)$

**3.** $f(x) = \sin(\ln x)$

**4.** $f(x) = \ln(\sin^2 x)$

**5.** $f(x) = \log_2(1 - 3x)$

**6.** $f(x) = \log_5(xe^x)$

**7.** $f(x) = \sqrt[5]{\ln x}$

**8.** $f(x) = \ln \sqrt[5]{x}$

**9.** $f(x) = \sin x \ln(5x)$

**10.** $f(t) = \dfrac{1 + \ln t}{1 - \ln t}$

**11.** $F(t) = \ln \dfrac{(2t + 1)^3}{(3t - 1)^4}$

**12.** $h(x) = \ln\left(x + \sqrt{x^2 - 1}\right)$

**13.** $g(x) = \ln\left(x\sqrt{x^2 - 1}\right)$

**14.** $F(y) = y \ln(1 + e^y)$

**15.** $f(u) = \dfrac{\ln u}{1 + \ln(2u)}$

**16.** $y = \dfrac{1}{\ln x}$

**17.** $y = \ln|2 - x - 5x^2|$

**18.** $H(z) = \ln \sqrt{\dfrac{a^2 - z^2}{a^2 + z^2}}$

**19.** $y = \ln(e^{-x} + xe^{-x})$

**20.** $y = [\ln(1 + e^x)]^2$

**21.** $y = 2x \log_{10}\sqrt{x}$

**22.** $y = \log_2(e^{-x}\cos \pi x)$

$$\frac{d}{dx}\left[x^{\sin(x)} + \ln(x^{\sqrt{x}})\right]$$

## HYPERBOLIC FUNCTIONS

**DEFINITION OF THE HYPERBOLIC FUNCTIONS**

$$\sinh x = \frac{e^x - e^{-x}}{2} \qquad \operatorname{csch} x = \frac{1}{\sinh x}$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \operatorname{sech} x = \frac{1}{\cosh x}$$

$$\tanh x = \frac{\sinh x}{\cosh x} \qquad \coth x = \frac{\cosh x}{\sinh x}$$

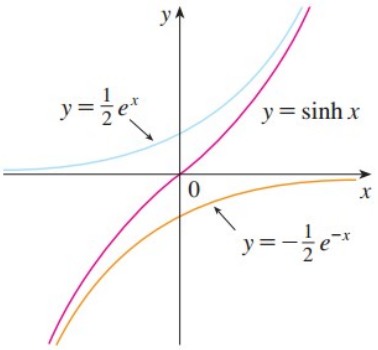



**FIGURE 1**
$y = \sinh x = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$

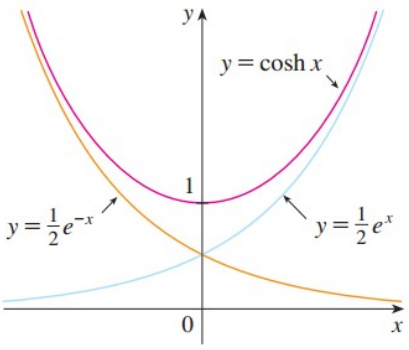



**FIGURE 2**
$y = \cosh x = \frac{1}{2}e^x + \frac{1}{2}e^{-x}$

**HYPERBOLIC IDENTITIES**

$$\sinh(-x) = -\sinh x \qquad \cosh(-x) = \cosh x$$

$$\cosh^2 x - \sinh^2 x = 1 \qquad 1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

**I DERIVATIVES OF HYPERBOLIC FUNCTIONS**

$$\frac{d}{dx}(\sinh x) = \cosh x \qquad \frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$

$$\frac{d}{dx}(\cosh x) = \sinh x \qquad \frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x \qquad \frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

**6 DERIVATIVES OF INVERSE HYPERBOLIC FUNCTIONS**

$$\frac{d}{dx}(\sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}} \qquad \frac{d}{dx}(\operatorname{csch}^{-1}x) = -\frac{1}{|x|\sqrt{x^2+1}}$$

$$\frac{d}{dx}(\cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}} \qquad \frac{d}{dx}(\operatorname{sech}^{-1}x) = -\frac{1}{x\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tanh^{-1}x) = \frac{1}{1-x^2} \qquad \frac{d}{dx}(\coth^{-1}x) = \frac{1}{1-x^2}$$

## 3.9 Linearization and Differentials

**DEFINITIONS** If $f$ is differentiable at $x = a$, then the approximating function

$$L(x) = f(a) + f'(a)(x - a)$$

is the **linearization** of $f$ at $a$. The approximation

$$f(x) \approx L(x)$$

of $f$ by $L$ is the **standard linear approximation** of $f$ at $a$. The point $x = a$ is the **center** of the approximation.

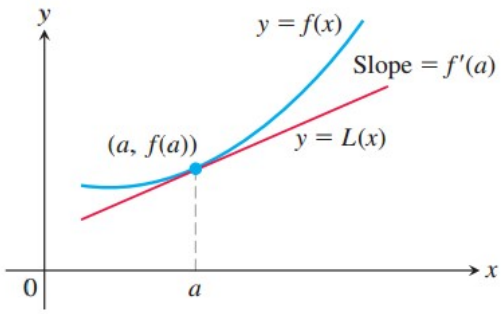



**FIGURE 3.40** The tangent to the curve $y = f(x)$ at $x = a$ is the line $L(x) = f(a) + f'(a)(x - a)$.

**EXAMPLE 1** Find the linearization of $f(x) = \sqrt{1 + x}$ at $x = 0$ (Figure 3.41).

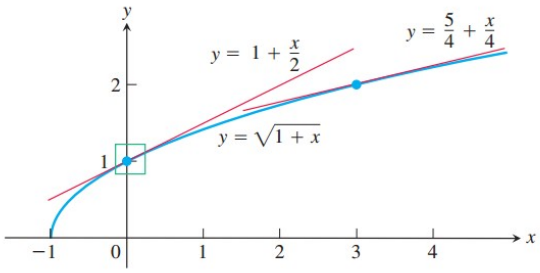



**FIGURE 3.41** The graph of $y = \sqrt{1 + x}$ and its linearizations at $x = 0$ and $x = 3$. Figure 3.42 shows a magnified view of the small window about 1 on the $y$-axis.

**EXAMPLE 1** Find the linearization of the function $f(x) = \sqrt{x + 3}$ at $a = 1$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$. Are these approximations overestimates or underestimates?

2 $$L(x) = f(a) + f'(a)(x - a)$$

is called the **linearization** of $f$ at $a$.

**EXAMPLE 3** Find the linearization of $f(x) = \cos x$ at $x = \pi/2$ (Figure 3.43).

## Differentials

We sometimes use the Leibniz notation $dy/dx$ to represent the derivative of $y$ with respect to $x$. Contrary to its appearance, it is not a ratio. We now introduce two new variables $dx$ and $dy$ with the property that when their ratio exists, it is equal to the derivative.

**DEFINITION** Let $y = f(x)$ be a differentiable function. The **differential *dx*** is an independent variable. The **differential *dy*** is

$$dy = f'(x)\,dx.$$

**EXAMPLE 4**

**(a)** Find $dy$ if $y = x^5 + 37x$.

**(b)** Find the value of $dy$ when $x = 1$ and $dx = 0.2$.

The geometric meaning of differentials is shown in Figure 3.44. Let $x = a$ and set $dx = \Delta x$. The corresponding change in $y = f(x)$ is

$$\Delta y = f(a + dx) - f(a).$$

The corresponding change in the tangent line $L$ is

$$\begin{aligned}\Delta L &= L(a + dx) - L(a) \\ &= \underbrace{f(a) + f'(a)[(a + dx) - a]}_{L(a + dx)} - \underbrace{f(a)}_{L(a)} \\ &= f'(a)\,dx.\end{aligned}$$

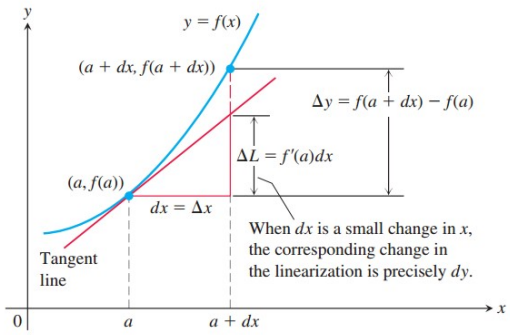



**FIGURE 3.44** Geometrically, the differential $dy$ is the change $\Delta L$ in the linearization of $f$ when $x = a$ changes by an amount $dx = \Delta x$.

**EXAMPLE 3** Compare the values of $\Delta y$ and $dy$ if $y = f(x) = x^3 + x^2 - 2x + 1$ and $x$ changes (a) from 2 to 2.05 and (b) from 2 to 2.01.

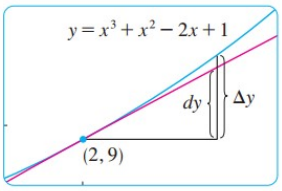



**FIGURE 6**

## Estimating with Differentials

Suppose we know the value of a differentiable function $f(x)$ at a point $a$ and want to estimate how much this value will change if we move to a nearby point $a + dx$. If $dx = \Delta x$ is small, then we can see from Figure 3.44 that $\Delta y$ is approximately equal to the differential $dy$. Since

$$f(a + dx) = f(a) + \Delta y, \qquad \Delta x = dx$$

the differential approximation gives

$$f(a + dx) \approx f(a) + dy$$

when $dx = \Delta x$. Thus the approximation $\Delta y \approx dy$ can be used to estimate $f(a + dx)$ when $f(a)$ is known, $dx$ is small, and $dy = f'(a)\,dx$.

**EXAMPLE 6** The radius $r$ of a circle increases from $a = 10$ m to 10.1 m (Figure 3.45). Use $dA$ to estimate the increase in the circle's area $A$. Estimate the area of the enlarged circle and compare your estimate to the true area found by direct calculation.

**EXAMPLE 7** Use differentials to estimate

**(a)** $7.97^{1/3}$ **(b)** $\sin(\pi/6 + 0.01)$.

## EXERCISES 3.9

### Finding Linearizations

In Exercises 1–5, find the linearization $L(x)$ of $f(x)$ at $x = a$.

1. $f(x) = x^3 - 2x + 3, \quad a = 2$
2. $f(x) = \sqrt{x^2 + 9}, \quad a = -4$
3. $f(x) = x + \frac{1}{x}, \quad a = 1$
4. $f(x) = \sqrt[3]{x}, \quad a = -8$
5. $f(x) = \tan x, \quad a = \pi$
6. **Common linear approximations at $x = 0$** Find the linearizations of the following functions at $x = 0$.
   **a.** $\sin x$ **b.** $\cos x$ **c.** $\tan x$

### Linearization for Approximation

In Exercises 7–12, find a linearization at a suitably chosen integer near $a$ at which the given function and its derivative are easy to evaluate.

7. $f(x) = x^2 + 2x, \quad a = 0.1$
8. $f(x) = x^{-1}, \quad a = 0.9$
9. $f(x) = 2x^2 + 3x - 3, \quad a = -0.9$
10. $f(x) = 1 + x, \quad a = 8.1$
11. $f(x) = \sqrt[3]{x}, \quad a = 8.5$
12. $f(x) = \frac{x}{x+1}, \quad a = 1.3$
13. Show that the linearization of $f(x) = (1 + x)^k$ at $x = 0$ is $L(x) = 1 + kx$.
14. Use the linear approximation $(1 + x)^k \approx 1 + kx$ to find an approximation for the function $f(x)$ for values of $x$ near zero.
    **a.** $f(x) = (1 - x)^6$ **b.** $f(x) = \frac{2}{1 - x}$
    **c.** $f(x) = \frac{1}{\sqrt{1 + x}}$ **d.** $f(x) = \sqrt{2 + x^2}$
    **e.** $f(x) = (4 + 3x)^{1/3}$ **f.** $f(x) = \sqrt[3]{\left(1 - \frac{x}{2 + x}\right)^2}$
15. **Faster than a calculator** Use the approximation $(1 + x)^k \approx 1 + kx$ to estimate the following.
    **a.** $(1.0002)^{50}$ **b.** $\sqrt[3]{1.009}$
16. Find the linearization of $f(x) = \sqrt{x + 1} + \sin x$ at $x = 0$. How is it related to the individual linearizations of $\sqrt{x + 1}$ and $\sin x$ at $x = 0$?

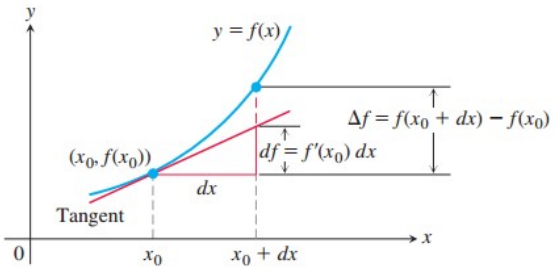
### Approximation Error

In Exercises 29–34, each function $f(x)$ changes value when $x$ changes from $x_0$ to $x_0 + dx$. Find

**a.** the change $\Delta f = f(x_0 + dx) - f(x_0)$;

**b.** the value of the estimate $df = f'(x_0)\,dx$; and

**c.** the approximation error $|\Delta f - df|$.




**29.** $f(x) = x^2 + 2x, \quad x_0 = 1, \quad dx = 0.1$

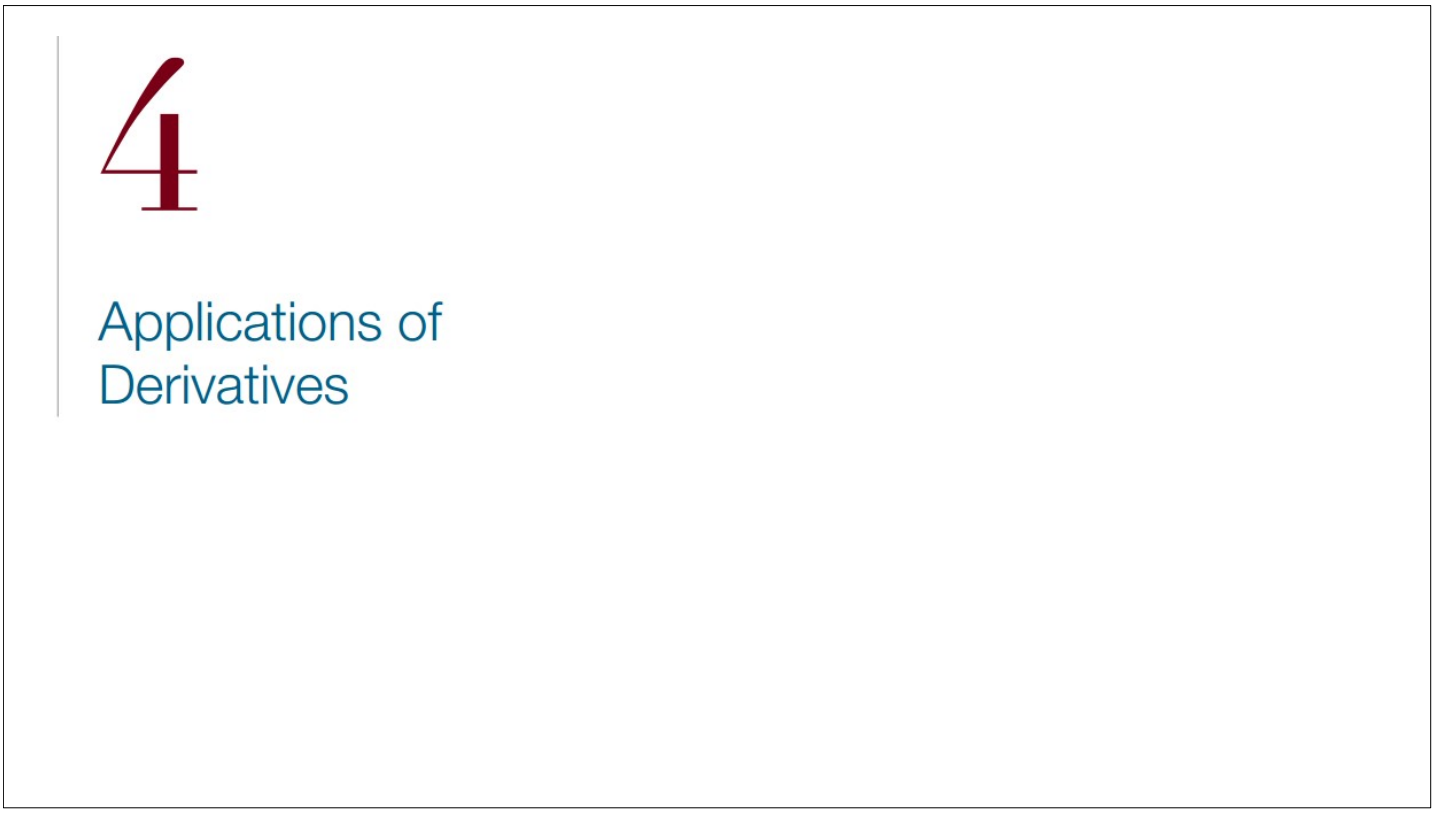
**30.** $f(x) = 2x^2 + 4x - 3, \quad x_0 = -1, \quad dx = 0.1$

**31.** $f(x) = x^3 - x, \quad x_0 = 1, \quad dx = 0.1$

**32.** $f(x) = x^4, \quad x_0 = 1, \quad dx = 0.1$

**33.** $f(x) = x^{-1}, \quad x_0 = 0.5, \quad dx = 0.1$

**34.** $f(x) = x^3 - 2x + 3, \quad x_0 = 2, \quad dx = 0.1$


4
Applications of
Derivatives

## 4.1 Extreme Values of Functions on Closed Intervals

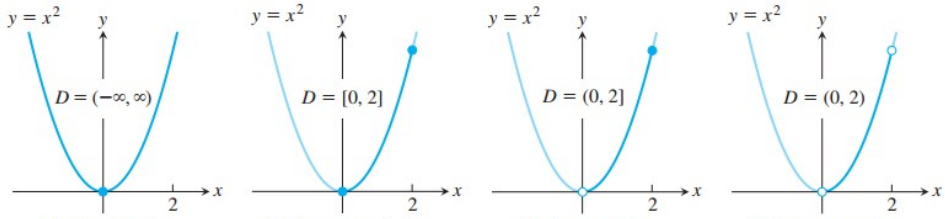
**DEFINITIONS** Let $f$ be a function with domain $D$. Then $f$ has an **absolute maximum** value on $D$ at a point $c$ if

$$f(x) \leq f(c) \qquad \text{for all } x \text{ in } D$$

and an **absolute minimum** value on $D$ at $c$ if

$$f(x) \geq f(c) \qquad \text{for all } x \text{ in } D.$$

| Function rule | Domain $D$ | Absolute extrema on $D$ |
|---|---|---|
| **(a)** $y = x^2$ | $(-\infty, \infty)$ | No absolute maximum<br>Absolute minimum of 0 at $x = 0$ |
| **(b)** $y = x^2$ | $[0, 2]$ | Absolute maximum of 4 at $x = 2$<br>Absolute minimum of 0 at $x = 0$ |
| **(c)** $y = x^2$ | $(0, 2]$ | Absolute maximum of 4 at $x = 2$<br>No absolute minimum |
| **(d)** $y = x^2$ | $(0, 2)$ | No absolute extrema |




(a) abs min only (b) abs max and min (c) abs max only (d) no max or min

**FIGURE 4.2** Graphs for Example 1.

**THEOREM 1—The Extreme Value Theorem**

If $f$ is continuous on a closed interval $[a, b]$, then $f$ attains both an absolute maximum value $M$ and an absolute minimum value $m$ in $[a, b]$. That is, there are numbers $x_1$ and $x_2$ in $[a, b]$ with $f(x_1) = m$, $f(x_2) = M$, and $m \leq f(x) \leq M$ for every other $x$ in $[a, b]$.

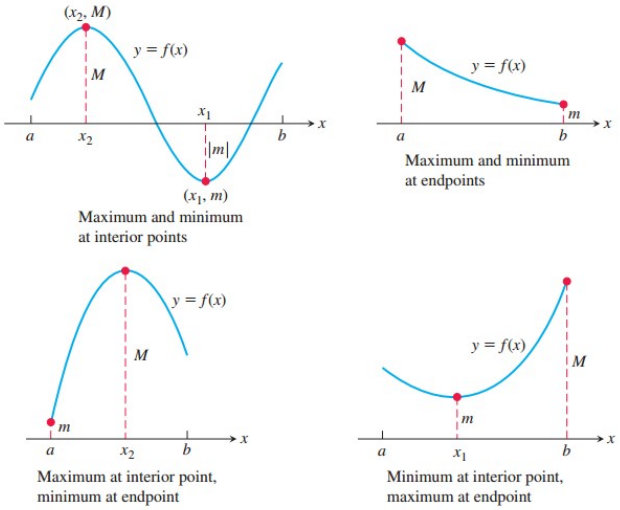



**FIGURE 4.3** Some possibilities for a continuous function's maximum and minimum on a closed interval $[a, b]$.

## Local (Relative) Extreme Values

Figure 4.5 shows a graph with five points where a function has extreme values on its domain $[a, b]$. The function's absolute minimum occurs at $a$ even though at $e$ the function's value is smaller than at any other point *nearby*. The curve rises to the left and falls to the right around $c$, making $f(c)$ a maximum locally. The function attains its absolute maximum at $d$. We now define what we mean by local extrema.

**DEFINITIONS** A function $f$ has a **local maximum** value at a point $c$ within its domain $D$ if $f(x) \leq f(c)$ for all $x \in D$ lying in some open interval containing $c$.

A function $f$ has a **local minimum** value at a point $c$ within its domain $D$ if $f(x) \geq f(c)$ for all $x \in D$ lying in some open interval containing $c$.

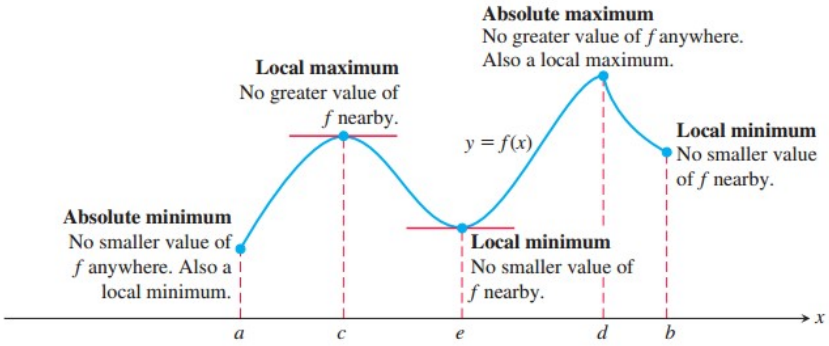



**FIGURE 4.5** How to identify types of maxima and minima for a function with domain $a \le x \le b$.

## Finding Extrema

The next theorem explains why we usually need to investigate only a few values to find a function's extrema.

**THEOREM 2—The First Derivative Theorem for Local Extreme Values**
If $f$ has a local maximum or minimum value at an interior point $c$ of its domain, and if $f'$ is defined at $c$, then

$$f'(c) = 0.$$

**DEFINITION** An interior point of the domain of a function $f$ where $f'$ is zero or undefined is a **critical point** of $f$.

**Finding the Absolute Extrema of a Continuous Function $f$ on a Finite Closed Interval**

1. Find all critical points of $f$ on the interval.
2. Evaluate $f$ at all critical points and endpoints.
3. Take the largest and smallest of these values.

**EXAMPLE 2** Find the absolute maximum and minimum values of $f(x) = x^2$ on $[-2, 1]$.

**7** If $f$ has a local maximum or minimum at $c$, then $c$ is a critical number of $f$.

**EXAMPLE 7** Find the critical numbers of $f(x) = x^{3/5}(4 - x)$.

**THE CLOSED INTERVAL METHOD** To find the *absolute* maximum and minimum values of a continuous function $f$ on a closed interval $[a, b]$:

1. Find the values of $f$ at the critical numbers of $f$ in $(a, b)$.
2. Find the values of $f$ at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

**EXAMPLE 8** Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1 \qquad -\tfrac{1}{2} \leq x \leq 4$$

**EXAMPLE 9**

(a) Use a graphing device to estimate the absolute minimum and maximum values of the function $f(x) = x - 2\sin x,\ 0 \leq x \leq 2\pi$.

(b) Use calculus to find the exact minimum and maximum values.

### Absolute Extrema on Finite Closed Intervals

In Exercises 21–36, find the absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

**21.** $f(x) = \frac{2}{3}x - 5, \quad -2 \le x \le 3$

**22.** $f(x) = -x - 4, \quad -4 \le x \le 1$

**23.** $f(x) = x^2 - 1, \quad -1 \le x \le 2$

**24.** $f(x) = 4 - x^3, \quad -2 \le x \le 1$

**25.** $F(x) = -\frac{1}{x^2}, \quad 0.5 \le x \le 2$

**26.** $F(x) = -\frac{1}{x}, \quad -2 \le x \le -1$

**27.** $h(x) = \sqrt[3]{x}, \quad -1 \le x \le 8$

**28.** $h(x) = -3x^{2/3}, \quad -1 \le x \le 1$

**29.** $g(x) = \sqrt{4 - x^2}, \quad -2 \le x \le 1$

**30.** $g(x) = -\sqrt{5 - x^2}, \quad -\sqrt{5} \le x \le 0$

**31.** $f(\theta) = \sin\theta, \quad -\frac{\pi}{2} \le \theta \le \frac{5\pi}{6}$